

CTU-Chat: Supervisor-Routed Multi-Agent RAG with Heterogeneous Knowledge Allocation for University Counseling

Ngoc-Anh Le^[0009-0006-7453-1009], Nhu-Quynh Nguyen^[0009-0008-7396-697X],
Tuong-Hung Cao^[0009-0003-5886-2666], Trang Minh Chanh, and Lan Phuong
Phan

College of Information and Communication Technology,
Can Tho University, Can Tho, Vietnam
{anhb2303861, quynhb2303777, hungb2303873, chanh}@student.ctu.edu.vn
pplan@ctu.edu.vn

Abstract. Higher education student counseling involves heterogeneous institutional knowledge, spanning relational curriculum graphs, cohort-specific tuition matrices, statutory exemption criteria, and narrative administrative decrees. Monolithic RAG frameworks and uncoordinated agent meshes can be brittle in this setting because of uniform representations, cross-domain evidence interference, and unconstrained tool exposure. We present **CTU-Chat**, a centralized supervisor-routed, domain-specialized multi-agent architecture in which each specialist is assigned a domain-specific knowledge space, retrieval path, and tool privilege for Vietnamese university counseling. Under centralized supervisor coordination, CTU-CHAT dispatches user queries to four bounded specialists with asymmetric privileges across three tiers: relational academic curricula are resolved through dedicated Cypher tools over a Neo4j knowledge graph, structured tuition rates are resolved via graph lookup and JSON catalogs with deterministic arithmetic engines, and narrative administrative regulations are retrieved through a hybrid document RAG path (BM25 and dense embeddings fused via Reciprocal Rank Fusion and cross-encoder reranking). Multi-agent evaluation demonstrates 97.0% supervisor routing accuracy (97.8% Macro-F1) and 100.0% bounded completion under strict invariants ($\text{max_tool_calls} = 1$, $\text{timeout} = 60.0\text{ s}$). Across 60 production-tool cases repeated three times, academic lookup, tuition-policy, scholarship, and no-tool gates achieved 100.0% end-to-end pass, while graph tuition lookup achieved 80.0% and graph exemption-basis lookup 73.3%. On the 150-question benchmark, progressive component stacking achieves 0.9000 Hit@1, 0.9600 Hit@3, 0.8656 Recall@5, and 0.9327 MRR.

Keywords: Multi-Agent Systems · Domain-Specialized Agents · Centralized Supervisor Routing · Heterogeneous Knowledge Allocation · Retrieval-Augmented Generation · University Counseling

1 Introduction

Providing timely, accessible, and accurate administrative guidance is vital for student success and institutional efficiency. In large universities such as Can Tho University (CTU)—a flagship institution in Vietnam’s Mekong Delta serving over 40,000 students [1]—advisory inquiries span diverse administrative domains. Because regulations are frequently updated through ministerial circulars, rector decisions, and cohort-specific guidelines, automated conversational counseling systems are essential to alleviate advisory strain and ensure compliance.

However, higher education counseling is characterized by acute *knowledge heterogeneity*, encompassing structurally distinct representations: (1) **Academic Curricula**: relational course prerequisite chains spanning the 113 majors represented in our graph corpus; (2) **Tuition Schedules**: multi-dimensional rate matrices governed by admission cohort and discipline; (3) **Statutory Exemptions**: legal decrees prescribing percentage deductions for qualified beneficiaries [6,7]; (4) **Scholarships**: multi-variable eligibility thresholds (cumulative GPA and training point scores); and (5) **Campus Regulations**: narrative administrative prose governing examinations, conduct, and student loans.

To automate institutional QA, Retrieval-Augmented Generation (RAG) [10,5] and GraphRAG [3] augment large language models (LLMs) with external documents or knowledge graphs, while agentic architectures [20,18,16] introduce iterative tool invocation and multi-agent coordination. Nevertheless, applying existing paradigms to university counseling reveals significant limitations. Monolithic RAG pipelines risk a *uniform representation fallacy*: forcing relational curricula, tabular fee matrices, and narrative legal prose into a single retrieval index can fragment tabular data and prerequisite links. Conversely, uncoordinated multi-agent meshes can expose broad tool sets across agents, increasing the risk of tool mis-selection, parameter errors, and extended execution, while known limitations in language-model arithmetic [14] motivate deterministic calculation for credit-based tuition.

These challenges crystallize three open research gaps in automated university counseling: (1) **Gap 1 (Limited Domain-Specialized Multi-Agent Orchestration)**: Existing agentic and multi-agent systems provide delegation and tool use, but the coordinated use of domain-specialized agents with isolated knowledge spaces, tool privileges, and retrieval paths remains underexplored in heterogeneous university counseling; (2) **Gap 2 (Uniform Knowledge Representation)**: Existing RAG-oriented systems often process structurally different institutional knowledge through a common retrieval space, although relational curricula, structured tuition information, and narrative regulations require fundamentally different representations; and (3) **Gap 3 (Lack of Representation-Aware Routing)**: Existing systems provide lexical, semantic, and graph retrieval methods, but offer limited mechanisms for selecting the appropriate evidence representation and specialist according to query domain and information structure.

To address these gaps, we formulate three core research questions:

- **RQ1 (Domain-Specialized Multi-Agent Architecture):** How effectively can a centralized supervisor-routed multi-agent architecture coordinate domain-specialized agents with isolated knowledge and tool access for heterogeneous university counseling tasks?
- **RQ2 (Heterogeneous Knowledge Allocation):** How does prioritizing graph-oriented access for relational academic-program and structured tuition knowledge, while retaining narrative regulations in hybrid document retrieval, affect retrieval and answer quality compared with uniform retrieval representations?
- **RQ3 (Representation-Aware Evidence Routing):** How does intent-guided routing between graph-oriented and hybrid document-retrieval paths affect retrieval effectiveness, answer correctness, and latency across different counseling domains?

We hypothesize that reliable university counseling is better supported by a supervisor-routed multi-agent architecture in which domain specialists are constrained to knowledge representations and tools appropriate to their information structure, rather than by a uniform retrieval-and-reasoning pipeline. Accordingly, CTU-CHAT separates specialist coordination, knowledge allocation, and evidence-path selection as three complementary design dimensions.

To answer these questions, we present **CTU-Chat**. CTU-CHAT *proposes a centralized supervisor-routed, domain-specialized multi-agent architecture in which each specialist is assigned a domain-specific knowledge space, retrieval path, and tool privilege*. Under centralized supervisor coordination, incoming inquiries are contextualized via Redis history and dispatched to four bounded specialists operating under asymmetric privileges: (1) **Academic Specialist**, bound to 6 dedicated Cypher tools over a Neo4j knowledge graph; (2) **Financial Specialist**, equipped with structured tuition graph lookups and the deterministic arithmetic engine `tin_h_toan_hoc_phi`; (3) **Scholarship Specialist**, equipped with the deterministic evaluator `tin_h_tien_hoc_bong`; and (4) **General Specialist**, operating with zero tools to synthesize narrative regulatory context. Specialists share session state in Redis while maintaining strictly isolated execution envelopes without unconstrained peer-to-peer communication.

Our principal contributions are three-fold:

- **C1 (Supervisor-Routed Domain-Specialized Multi-Agent Architecture):** A centralized supervisor-routed architecture coordinating four domain-specialized agents with asymmetric tool privileges, a per-decision bound of `max_tool_calls = 1`, and shared session state, designed to reduce tool mis-selection and constrain execution without peer-to-peer delegation (addressing Gap 1 and RQ1).
- **C2 (Heterogeneous Knowledge Allocation Framework):** A principled allocation strategy that separates relational curricula and structured tuition into graph-oriented representations while reserving hybrid document retrieval for narrative administrative decrees, resolving the uniform representation fallacy (addressing Gap 2 and RQ2).

- **C3 (Representation-Aware Routing and Evidence Acquisition):** An intent-guided evidence acquisition mechanism that routes queries dynamically to graph-oriented or hybrid document-retrieval paths, leveraging established BM25, Dense vectors, RRF, and cross-encoder reranking as complementary building blocks (addressing Gap 3 and RQ3).

2 Related Work

This section reviews established retrieval and reasoning paradigms in the literature, examining their operational mechanisms and limitations in institutional counseling.

2.1 Sparse and Dense Retrieval

Lexical Retrieval (BM25): Sparse keyword search using BM25 [15] evaluates term frequency (tf) and inverse document frequency (IDF) over an inverted index:

$$s_{\text{BM25}}(q, d) = \sum_{w \in q} \text{IDF}(w) \cdot \frac{\text{tf}(w, d) \cdot (k_1 + 1)}{\text{tf}(w, d) + k_1 \cdot \left(1 - b + b \cdot \frac{|d|}{|d|}\right)}, \quad (1)$$

with $k_1 = 1.5$ and $b = 0.75$. BM25 supports exact lexical matching for rigid alphanumeric tokens (decision codes, cohort labels, course identifiers), but may miss semantic paraphrases.

Dense Semantic Retrieval: Dense retrieval systems [9,10] map queries and passages into embedding spaces using bi-encoders, computing relevance via vector similarity. While dense models capture paraphrastic intent, rare alphanumeric identifiers and out-of-vocabulary regulatory tokens can remain difficult and may introduce cross-cohort noise, motivating hybrid retrieval.

2.2 Hybrid Retrieval and Reciprocal Rank Fusion

To merge BM25 and dense candidates without score calibration, Reciprocal Rank Fusion (RRF) [2] sorts documents by reciprocal position: $s_{\text{RRF}}(d) = \sum_{s \in \mathcal{M}} \frac{1}{k + r_s(d)}$ ($k = 60$), while neural cross-encoders [12,19] jointly encode query–passage pairs for reranking. Adaptive-RAG further demonstrates that retrieval strategies can be selected according to question complexity [8]. These approaches motivate dynamic retrieval, but do not directly allocate institution-specific relational, tabular, and narrative knowledge to isolated specialist paths.

2.3 Knowledge Graphs and GraphRAG

Knowledge graphs represent domain knowledge as structured triples (h, r, t) [13]. GraphRAG [3] leverages graph topology to uncover multi-hop associative paths escaping flat chunking, benefiting curriculum structures and prerequisite chains. However, universal graph extraction across institutional corpora produces hallucinated relationships, while forcing narrative legal prose into graphs is structurally unnatural, motivating selective rather than universal graph use.

2.4 Agentic RAG and Multi-Agent Orchestration

ReAct demonstrates interleaved reasoning and action, AutoGen provides a framework for multi-agent conversation, and Toolformer studies when and how language models invoke external APIs [20,18,16]. For institutional counseling, these capabilities raise design risks when applied without explicit constraints: broad tool exposure can increase tool and parameter errors; peer-to-peer exchanges can increase coordination cost; a shared index can mix evidence across domains; and a common retrieval path can obscure opportunities for direct graph traversal or structured lookup. These risks motivate the bounded, supervisor-routed specialization evaluated in this work.

2.5 Educational and University Counseling Systems

Systematic evidence shows that educational chatbots are used for informational, assisting, mentoring, and learning-support roles, while evaluation quality and adaptation remain open challenges [17]. REBot is a recent example of applying semantically enriched RAG and graph routing to educational counseling [11]. However, this line of work does not establish that a single retrieval representation or generative arithmetic is appropriate for every counseling domain. We therefore study explicit representation allocation, specialist routing, and deterministic calculation as separate design dimensions.

2.6 Comparative Synthesis and Research Gaps

Synthesizing established paradigms reveals distinct structural trade-offs for our setting: (1) *BM25* [15] favors exact administrative tokens but may miss natural paraphrases; (2) *Dense Retrieval* [9] captures semantic intent but may confuse similar cohort-specific passages; (3) *Hybrid RRF* [2] combines lexical and semantic rankings but does not itself model relational curricula; (4) *GraphRAG* [3] supports graph-based associations, although universal graph extraction is not necessarily suitable for narrative regulations; (5) *Adaptive RAG* [8] selects retrieval depth by query complexity but not by institutional representation type; (6) *ReAct/Toolformer-style systems* [20,16] support tool-using reasoning but require tool-access controls; and (7) *multi-agent conversation frameworks* [18] provide coordination primitives but do not by themselves guarantee bounded execution or knowledge isolation.

This comparative analysis directly crystallizes the three core research gaps identified in Section 1: (1) **Gap 1 (Limited Domain-Specialized Multi-Agent Orchestration)**: Coordinated multi-agent systems with isolated knowledge spaces, tool privileges, and dedicated retrieval paths remain underexplored in university counseling; (2) **Gap 2 (Uniform Knowledge Representation)**: Processing relational curricula, structured tuition, and narrative regulations through a common retrieval space induces representation failure; and (3) **Gap 3 (Lack of Representation-Aware Routing)**: Dynamic routing mechanisms that steer queries to appropriate evidence representations and bounded specialists according to information structure are absent in existing architectures.

3 Proposed Model

This section presents the architectural design, algorithmic foundation, and operational mechanisms of CTU-CHAT. Unlike conventional monolithic RAG pipelines that force all institutional documentation into a uniform document index or graph schema, CTU-CHAT *proposes a centralized supervisor-routed, domain-specialized multi-agent architecture in which each specialist is assigned a domain-specific knowledge space, retrieval path, and tool privilege.*

3.1 Design Motivation and Knowledge Characteristics

Higher education counseling at large institutions such as Can Tho University (CTU) is characterized by acute knowledge heterogeneity. An empirical audit of 244 official institutional documents reveals five structurally distinct categories of administrative information, each imposing unique representation and reasoning requirements:

1. **Academic Curricula (Relational Knowledge):** Encompassing 113 undergraduate majors (alongside an institutional academic regulation document), curricula exhibit deep hierarchies (General → Fundamental → Specialized → Electives) and strict multi-hop prerequisite dependencies. While flat document chunking fragments prerequisite chains, a property graph naturally captures topological relationships, enabling deterministic path traversals without relation hallucination.
2. **Tuition Fee Schedules (Structured / Parametric Knowledge):** Governed by official institutional rector decisions, tuition fee schedules are multi-dimensional matrices indexed by admission cohorts and major groups. Flat text search frequently conflates cohorts, while generative LLMs miscalculate credit multipliers, requiring structured tabular lookups coupled with deterministic arithmetic engines.
3. **Statutory Fee Reductions (Normative Legal Decrees):** Governed by Decree 81/2021/ND-CP and its amendment, Decree 97/2023/ND-CP [6,7], statutory exemptions prescribe exact deductions for qualified beneficiaries. These require parent-child hybrid text retrieval for legal qualification, followed by exact mathematical deduction against student tuition.
4. **Scholarships and Financial Aid (Rule-Based Threshold Policies):** Academic encouragement scholarships operate under dual thresholds (cumulative GPA and training point scores) evaluated across tiers (Good, Very Good, Excellent) against semester quotas, combining rule-based filtering with narrative conditions.
5. **General Campus Regulations (Unstructured Administrative Prose):** Administrative decisions governing student conduct, examination retakes, dormitories, and deferrals are long narrative prose requiring high-recall semantic search paired with exact lexical matching for official decision identifiers and administrative codes.

These heterogeneous characteristics expose the fundamental flaw of uniform RAG systems: no single retrieval or reasoning paradigm is optimal for all administrative domains. This motivates our proposed architecture: coordinating domain-specialized agents with isolated knowledge spaces and asymmetric tool privileges under a centralized supervisor.

3.2 Overall Proposed Architecture

Figure 1 depicts the system architecture of CTU-CHAT, structured across three coordinated tiers: (1) the *Interaction and Centralized Supervisor Tier*, (2) the *Domain-Specialized Multi-Agent Core Layer*, and (3) the *Heterogeneous Knowledge Storage and Tool Tier*.

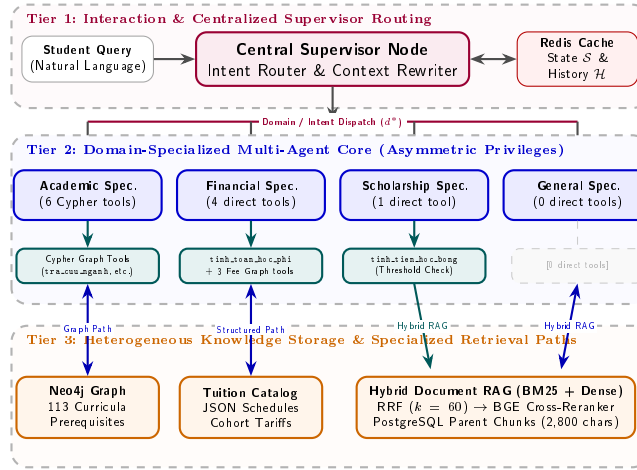


Fig. 1. System architecture of CTU-CHAT, highlighting the centralized supervisor, the domain-specialized multi-agent core layer, and separate knowledge paths for relational, structured, and narrative administrative evidence.

The interaction flow proceeds through three well-defined stages: (1) **Tier 1 (Supervisor)**: Student query q is contextualized against Redis history $\mathcal{H}$ and routed to predicted domain $d^* \in \mathcal{D}$; (2) **Tier 2 (Domain Multi-Agent Core)**: The query is dispatched to the designated specialist operating under asymmetric tool privileges (Table 1); and (3) **Tier 3 (Heterogeneous Knowledge Paths)**: The specialist queries isolated representations (Neo4j for curricula, structured graph/catalogs for fee matrices, and hybrid document RAG for narrative regulations) and executes bounded deterministic tools.

3.3 Supervisor-Routed Domain-Specialized Multi-Agent Architecture

A core contribution of CTU-CHAT is the operational isolation of domain agents within a **centralized supervisor-routed multi-agent architecture**. In conventional multi-agent meshes, exposing broad tool suites to all agents and permitting autonomous agent-to-agent negotiation can increase the risk of parameter

errors, cascading failures, and extended execution. CTU-CHAT is designed to reduce these risks through explicit structural boundaries:

- **Supervisor Agent (Central Dispatcher):** Implemented via a structured output LLM (`gemini-2.5-flash-lite` in empirical evaluation, configurable to `gemini-3.5-flash-lite` in operational deployments) with deterministic regex fallbacks. The supervisor inspects incoming query q and session history $\mathcal{H}$, generates a disambiguated standalone query $q_{\text{standalone}}$, classifies intent into exactly one target domain $d^* \in \{\text{academic, financial, scholarship, general}\}$, and selects the appropriate evidence acquisition path.
- **Four Domain Specialists:**
 1. **Academic Specialist:** Dedicated to curriculum structures across 113 undergraduate majors, prerequisite course dependency chains, and course equivalence. Bounded to 6 direct Cypher inspection tools connecting to Neo4j, with zero arithmetic tools and zero document retrieval tools. Reasoning mode: ReAct loop over graph tools.
 2. **Financial Specialist:** Dedicated to credit-based tuition fee schedules, cohort tariffs, statutory fee deductions, and payment deadlines. Armed with 4 direct tools including fee graph queries and the deterministic arithmetic engine `tin_h_toan_hoc_phi`.
 3. **Scholarship Specialist:** Dedicated to academic encouragement scholarships, corporate sponsored grants, and financial aid quotas. Armed with the deterministic threshold evaluation tool `tin_h_tien_hoc_bong`.
 4. **General Specialist:** Dedicated to administrative regulations, student conduct, examination retakes, dormitories, student loans, and pedagogical support. Operates with zero direct tools, synthesizing answers purely over retrieved hybrid parent-child regulatory passages.
- **Shared State vs. Isolated Resources:** All agents share conversational session state $\mathcal{S}$ and dialogue history $\mathcal{H}$ managed in Redis. However, tool namespaces, prompt envelopes, and knowledge stores are strictly isolated: no specialist has access to cross-domain tools or unauthorized databases.
- **Strict Centralized Routing (No Peer-to-Peer Communication):** Unlike collaborative agent meshes, CTU-CHAT strictly prohibits peer-to-peer handoffs, autonomous agent-to-agent negotiations, or dynamic delegation. All interactions follow the centralized routing topology: $\text{START} \rightarrow \text{Supervisor} \rightarrow [\text{Evidence Node}] \rightarrow \text{Specialist} \rightarrow \text{END}$.

Table 1. Operational comparison and resource privileges of domain-specialized agents in CTU-CHAT.

Specialist Agent	Shared State	Available Direct Tools	Primary Evidence Store	Retrieval Strategy	Tool-Call Bound
Academic Specialist	Redis $\mathcal{S}, \mathcal{H}$	6 Cypher tools (<code>tra_cus_nghanh</code> , etc.)	Neo4j Knowledge Graph	Direct Cypher topological traversal	<code>max_tool_calls = 1</code>
Financial Specialist	Redis $\mathcal{S}, \mathcal{H}$	4 tools (<code>tin_h_toan_hoc_phi</code> , etc.)	Tuition Graph + JSON Catalog	Pre-retrieved Fee Graph/JSON + Calc	<code>max_tool_calls = 1</code>
Scholarship Specialist	Redis $\mathcal{S}, \mathcal{H}$	1 tool (<code>tin_h_tien_hoc_bong</code>)	Hybrid RAG + Criteria Catalog	Pre-retrieved Hybrid + Criteria tool	<code>max_tool_calls = 1</code>
General Specialist	Redis $\mathcal{S}, \mathcal{H}$	0 direct tools (pure synthesis)	PostgreSQL + Qdrant	Parent-child Hybrid RRF + BGE	<code>max_tool_calls = 0</code>

Distinction Between Direct Tools and Pre-Retrieved Context: Table 1 formalizes the boundary between *direct tools bound to specialists* and

pre-retrieval functions executed outside the specialist. Specifically, the **Academic Specialist** executes its 6 Cypher tools directly during ReAct execution without passing through the text retrieval node. In contrast, for the **Financial Specialist** and **Scholarship Specialist**, structured tuition tables and hybrid regulatory passages are pre-retrieved and formatted into prompt context by the upstream retrieval node; direct tools are then invoked solely for deterministic calculations or interactive sub-queries.

3.4 Heterogeneous Knowledge Allocation

Rather than forcing all university data into a single retrieval index, CTU-CHAT allocates knowledge across specialized modalities according to structural properties (Table 2).

Table 2. Heterogeneous knowledge allocation across institutional administrative domains.

Knowledge Type	Structural Property	Primary Representation	Runtime Access Mechanism	Specialist
Academic Programs	Relational, multi-hop	Neo4j Knowledge Graph	Dedicated Cypher query tools	Academic
Tuition Schedules	Structured, cohort-dependent	Neo4j Tuition Graph / JSON	Graph lookup + deterministic calc	Financial
Scholarship Criteria	Narrative + rule-based thresholds	Hybrid Document RAG	Hybrid search + threshold tool	Scholarship
General Regulations	Narrative administrative text	Hybrid Document RAG	BM25 + Dense + RRF + BGE	General

Physical Storage vs. Primary Runtime Retrieval Path: We draw a crucial architectural distinction between physical indexing and runtime retrieval:

- *Physical Storage/Indexing:* Neo4j 5 hosts topological curriculum entities (**Major**, **KnowledgeBlock**, **Course**) and structured tuition nodes. Structured JSON files store frozen credit tariffs and scholarship thresholds. Simultaneously, 244 institutional regulatory decrees are archived in PostgreSQL (parent chunks, 2,800 characters) and Qdrant (child chunks, 400 characters).
- *Primary Runtime Retrieval Path:* At runtime, academic inquiries bypass text search entirely to query Neo4j directly via Cypher tools. For tuition inquiries, the primary runtime path executes structured graph/catalog lookup; document retrieval is only invoked for textual decrees (e.g., payment procedures). For scholarship and general queries, the runtime path executes hybrid document retrieval.

3.5 Representation-Aware Routing Strategy

Applying global hybrid retrieval across an undifferentiated administrative corpus introduces severe cross-domain evidence interference. To ensure precision, CTU-CHAT enforces rigorous decision criteria mapping inquiry characteristics to preferred evidence mechanisms (Table 3).

Formally, the supervisor maps incoming student query q and dialogue context $\mathcal{H}$ to a constrained routing tuple:

$$\mathcal{R}(q, \mathcal{H}) = \langle q_{\text{standalone}}, d^*, A^*, \mathcal{P}_{A^*}, \mathcal{L}_{\text{target}} \rangle, \quad (2)$$

where $q_{\text{standalone}}$ is the disambiguated query, $d^* \in \mathcal{D}$ denotes the predicted domain ($\mathcal{D} = \{\text{acad}, \text{fin}, \text{sch}, \text{gen}\}$), A^* is the assigned specialist agent, $\mathcal{P}_{A^*}$ defines

Table 3. Decision criteria for evidence representation and retrieval mechanism selection.

Information Characteristic	Preferred Mechanism	Structural Rationale
Exact identifiers, course codes, formal terms	BM25 Lexical Search	Exact inverted index token matching without semantic drift
Semantic paraphrase & conversational phrasing	Dense Vector Search (Qdrant)	Captures contextual proximity in 768-d embedding space
Narrative administrative policies & decrees	Hybrid BM25 + Dense + RRF + BGE	Fuses lexical and semantic signals, refined by cross-attention
Relational curriculum / prerequisite query	Neo4j Knowledge Graph	Deterministic graph traversal over topological dependencies
Statutory tuition / scholarship calculation	Structured Graph/JSON + Tools	Deterministic arithmetic outside generative text production

authorized tool privileges, and $\mathcal{L}_{\text{target}} \in \{\text{'direct_graph'}, \text{'evidence_node'}\}$ specifies the evidence acquisition path.

After identifying the need for contextualization, the supervisor jointly determines the target specialist and its corresponding evidence path. Algorithm 1 formalizes this centralized routing process. Given the current student query q , dialogue history $\mathcal{H}$, and shared session state $\mathcal{S}$, the supervisor optionally rewrites context-dependent queries, classifies domain and intent, assigns a bounded specialist, and selects the appropriate graph-oriented or document-oriented evidence path.

Algorithm 1: Supervisor Intent and Specialist Routing

Input: Incoming student query q , Session dialogue history $\mathcal{H}$, Shared session state $\mathcal{S}$
Output: Routing tuple $\langle q_{\text{standalone}}, d^*, A^*, \mathcal{P}_{A^*}, \mathcal{L}_{\text{target}} \rangle$

```

1  $q_{\text{standalone}} \leftarrow q$ ;
2 if ShouldRewriteQuery( $q, \mathcal{H}$ ) then
3    $q_{\text{cand}} \leftarrow \text{FastLLM\_Rewrite}(q, \mathcal{H})$ ;
4   if ValidateRewrite( $q, q_{\text{cand}}, \mathcal{H}$ ) then
5      $q_{\text{standalone}} \leftarrow q_{\text{cand}}$ ;
6  $\langle d^*, \iota \rangle \leftarrow \text{ClassifyDomainAndIntent}(q_{\text{standalone}}, \text{SUPERVISOR\_PROMPT})$ ;
7 if  $d^* = \text{NULL}$  then
8    $d^* \leftarrow \text{RegexFallbackClassifier}(q_{\text{standalone}})$ ;
9  $A^* \leftarrow \text{SelectSpecialist}(d^*)$ ;  $\mathcal{P}_{A^*} \leftarrow \text{GetSpecialistPrivileges}(A^*)$ ;
10 if  $d^* = \text{'academic'}$  then
11    $\mathcal{L}_{\text{target}} \leftarrow \text{'direct\_graph'}$ ;
12 else
13    $\mathcal{L}_{\text{target}} \leftarrow \text{'evidence\_node'}$ ;
14 return  $\langle q_{\text{standalone}}, d^*, A^*, \mathcal{P}_{A^*}, \mathcal{L}_{\text{target}} \rangle$ ;

```

The output of Algorithm 1 is not the final answer but a constrained routing decision tuple. This decision limits downstream execution strictly to the selected specialist A^* , designated evidence source $\mathcal{L}_{\text{target}}$, and permitted tool set $\mathcal{P}_{A^*}$, thereby enforcing the resource isolation that defines the proposed multi-agent architecture. By pruning unauthorized tools prior to specialist invocation, this centralized routing mechanism constrains tool exposure and execution paths; Scenario 3 evaluates the resulting bounded behavior at the isolated decision level.

3.6 Hybrid Document Retrieval Path

For narrative administrative regulations and scholarship policies, CTU-CHAT employs a multi-stage hybrid document retrieval pipeline:

1. **Dual Candidate Generation:** Query $q_{\text{standalone}}$ is searched simultaneously via BM25 over tokenized child chunks in PostgreSQL and dense vector search

in Qdrant (768-d embeddings via `vietnamese-bi-encoder`), returning candidate sets $\mathcal{D}_{\text{bm25}}$ and $\mathcal{D}_{\text{dense}}$ (top-15 in production defaults, with $k = 10$ evaluated in retrieval benchmarks).

2. **Reciprocal Rank Fusion (RRF)**: Candidates are merged without fragile score normalization:

$$s_{\text{RRF}}(d) = \sum_{s \in \{\text{bm25}, \text{dense}\}} \frac{1}{k + r_s(d)}, \quad k = 60. \quad (3)$$

3. **Neural Cross-Encoder Reranking**: The top-fused candidates are reranked via `BAAI/bge-reranker-v2-m3` cross-attention, selecting the top-5 child chunks (with top-6 configured as the engine production ceiling).
4. **Parent Chunk Expansion**: Selected child chunks are mapped back to their 2,800-character parent chunks in PostgreSQL, ensuring complete legal context and article structure for generation.

Clarification of Component Novelty: We emphasize that BM25, dense bi-encoders, RRF, and cross-encoders are established literature paradigms. In CTU-CHAT, they function as building blocks within the hybrid path, coordinated under supervisor routing.

3.7 Graph-Oriented Academic and Tuition Retrieval

Curriculum Knowledge Graph (Neo4j): Curricula across all 113 undergraduate programs are modeled in Neo4j (alongside general academic regulations). The **Academic Specialist** queries this graph directly via 6 APOC-optimized Cypher tools, resolving complex multi-hop prerequisite chains deterministically without relation hallucination.

Tuition Knowledge Graph and Structured Catalogs: Credit-based tuition fee schedules are indexed as structured nodes in Neo4j with a JSON catalog fallback. For statutory calculations, parameter extraction invokes the deterministic engine `tinhh.toan.hoc.ph`:

$$\text{Gross} = \sum_{i \in \mathcal{C}} c_i \cdot r_i \cdot m_i, \quad (4)$$

$$\text{Reduction} = \sum_{i \in \mathcal{C}} c_i \cdot b_i \cdot (p_i/100), \quad (5)$$

$$\text{Payable} = \max(0, \text{Round}_{\text{VND}}(\text{Gross} - \text{Reduction})), \quad (6)$$

where $\mathcal{C}$ is the registered course set, c_i is course credit count, r_i is per-credit fee, m_i is cohort multiplier, b_i is exemption base rate, and $p_i \in \{0, 50, 70, 100\}$ is statutory deduction percentage. A typed Pydantic tool schema ensures numerical reliability.

3.8 Domain-Aware Evidence Acquisition and Specialist Execution

Algorithm 2 executes the routing decision produced by Algorithm 1. Evidence acquisition is intentionally asymmetric across specialists: academic queries are resolved through graph-oriented evidence, financial queries use structured tuition

evidence and deterministic calculation when required, while narrative administrative queries are served through hybrid document retrieval. This separation is central to CTU-CHAT and distinguishes it from a uniform retrieve-then-generate pipeline.

Algorithm 2: Evidence Acquisition and Bounded Specialist Execution

Input: Routing tuple $\langle q_{\text{standalone}}, d^*, A^*, \mathcal{P}_{A^*}, \mathcal{L}_{\text{target}} \rangle$, Knowledge stores (Neo4j, Qdrant, PostgreSQL, JSON)

Output: Validated citation-grounded response y_{final}

```

1 if  $A^* = \text{AcademicSpecialist}$  then
2    $\mathcal{E}_{\text{graph}} \leftarrow \text{AcquireGraphEvidence}(A^*, \mathcal{P}_{A^*}.\text{tools}, q_{\text{standalone}})$ ;
3    $y \leftarrow \text{SynthesizeSpecialistResponse}(A^*, q_{\text{standalone}}, \mathcal{E}_{\text{graph}})$ ;
4 else
5   switch  $d^*$  do
6     case 'financial' do
7        $\mathcal{E}_{\text{struct}} \leftarrow \text{LookupStructuredTuition}(q_{\text{standalone}})$ ;
8        $\mathcal{E}_{\text{policy}} \leftarrow \text{RetrieveRegulatoryDecrees}(q_{\text{standalone}})$ ;
9        $\mathcal{C}_{\text{ctx}} \leftarrow \mathcal{E}_{\text{struct}} \cup \mathcal{E}_{\text{policy}}$ ;
10      if  $\text{IsCalculationRequired}(q_{\text{standalone}})$  then
11         $\text{params} \leftarrow \text{ExtractFeeParameters}(q_{\text{standalone}}, \mathcal{C}_{\text{ctx}})$ ;
12         $\text{calc} \leftarrow \text{ExecuteDeterministicTool}(\text{tinh\_toan\_hoc\_phi}, \text{params})$ ;
13         $y \leftarrow \text{SynthesizeSpecialistResponse}(A^*, q_{\text{standalone}}, \mathcal{C}_{\text{ctx}} \cup \{\text{calc}\})$ ;
14      else
15         $y \leftarrow \text{SynthesizeSpecialistResponse}(A^*, q_{\text{standalone}}, \mathcal{C}_{\text{ctx}})$ ;
16     case 'scholarship' do
17        $\mathcal{E}_{\text{sch}} \leftarrow \text{RetrieveScholarshipEvidence}(q_{\text{standalone}})$ ;
18       if  $\text{IsThresholdCheckRequired}(q_{\text{standalone}})$  then
19         $\text{params} \leftarrow \text{ExtractCriteriaParameters}(q_{\text{standalone}}, \mathcal{E}_{\text{sch}})$ ;
20         $\text{eval} \leftarrow \text{ExecuteDeterministicTool}(\text{tinh\_tien\_hoc\_bong}, \text{params})$ ;
21         $y \leftarrow \text{SynthesizeSpecialistResponse}(A^*, q_{\text{standalone}}, \mathcal{E}_{\text{sch}} \cup \{\text{eval}\})$ ;
22      else
23         $y \leftarrow \text{SynthesizeSpecialistResponse}(A^*, q_{\text{standalone}}, \mathcal{E}_{\text{sch}})$ ;
24     case 'general' do
25        $\mathcal{E}_{\text{gen}} \leftarrow \text{RetrieveHybridDocumentEvidence}(q_{\text{standalone}})$ ;
26        $y \leftarrow \text{SynthesizeGeneralResponse}(A^*, q_{\text{standalone}}, \mathcal{E}_{\text{gen}})$ ;
27  $y_{\text{final}} \leftarrow \text{AppendAuthoritativeCitations}(y)$ ;
28  $\text{CommitSessionState}(\text{Redis}, \mathcal{S}, q, y_{\text{final}})$ ;
29 return  $y_{\text{final}}$ ;

```

Algorithm 2 underscores the operational differentiation and privilege boundaries among the four specialists:

- **Academic vs. General Specialists:** The Academic Specialist queries a topological property graph (Neo4j) via a bounded ReAct loop over 6 Cypher tools to resolve multi-hop course dependencies, whereas the General Specialist operates with zero direct tools, synthesizing answers strictly over 2,800-character parent regulatory chunks retrieved through the hybrid RAG path.
- **Financial vs. Scholarship Specialists:** The Financial Specialist accesses structured fee schedules in Neo4j and JSON catalogs, executing the deterministic arithmetic engine `tinh_toan_hoc_phi` for exact monetary deductions. In contrast, the Scholarship Specialist retrieves narrative eligibility decrees and invokes `tinh_tien_hoc_bong` to evaluate discrete GPA and training-point thresholds against semester quotas.

- **Specialist Knowledge Paths and Tools:** Academic queries utilize the graph path exclusively; financial queries prioritize structured graph/JSON lookups paired with deterministic arithmetic; scholarship and general inquiries rely on the multi-stage hybrid document RAG path (BM25, Dense embeddings, RRF, and neural cross-encoder reranking).
- **Shared State vs. Isolated Resources:** While dialogue history $\mathcal{H}$ and session state $\mathcal{S}$ are shared via Redis, each specialist is strictly bound to its own prompt envelope, knowledge store, and authorized toolset. Peer-to-peer delegation is prohibited, with at most one tool call permitted per evaluated decision (`max_tool_calls = 1`).

The key distinction is not the individual retrieval algorithms themselves, but the specialist-specific assignment of evidence paths and tool privileges. By pairing domain specialists with native data representations and deterministic engines, CTU-CHAT replaces fragile generative guesswork with reliable, verified execution.

3.9 Detailed Activity Flow

Figure 2 illustrates the runtime activity flowchart of CTU-CHAT, demonstrating the operational coordination between the central supervisor and the four domain-specialized execution swimlanes.

End-to-End Execution Walkthrough: As illustrated in Figure 2, a relational curriculum query is routed by the supervisor to the **Academic Specialist** and resolved through graph evidence, whereas a narrative regulation query is directed to the **General Specialist** and processed through the hybrid document-retrieval path. Financial queries additionally activate structured evidence lookup and deterministic calculation when the requested answer requires an exact monetary result, reducing reliance on generative arithmetic. Finally, retrieved regulatory citations are appended and conversational state is committed to Redis before returning the response to the student.

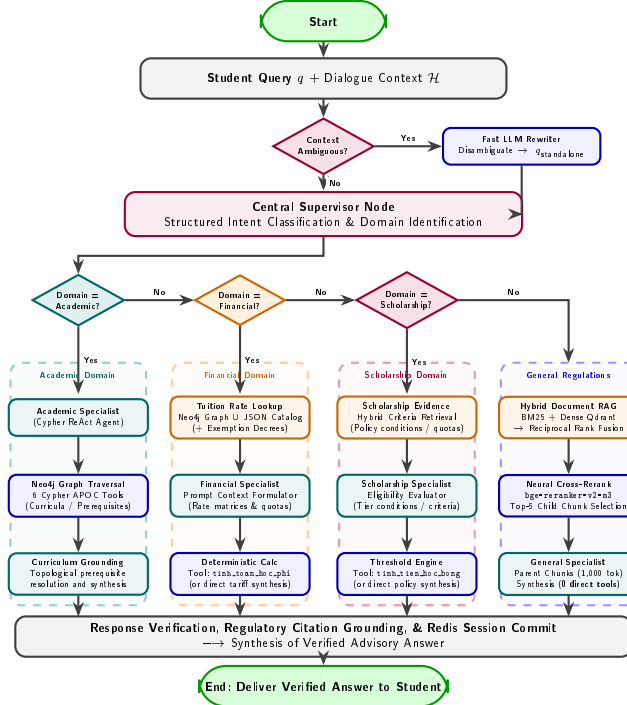


Fig. 2. Detailed runtime activity flowchart of CTU-CHAT, illustrating explicit Start/End states, decision diamonds with Yes/No branches for contextual query rewriting and cascading domain intent routing across four isolated specialist swimlanes, heterogeneous knowledge acquisition, bounded tool execution, and authoritative citation grounding.

4 Experimental Results

This section presents the empirical evaluation of CTU-CHAT, detailing the benchmark dataset, infrastructure, evaluation metrics, three experimental scenarios, discussion answering each research question in order, and threats to validity.

4.1 Experimental Dataset and Benchmark Construction

Scenarios 1 and 2 use the fixed file `data/150_NATURAL_NO_APPENDIX.csv`, a 150-question benchmark covering tuition fees, academic regulations, scholarships, and exemption policies. To construct it, we collected publicly available official university documents (including regulations, tuition schedules, scholarship and exemption policies, and academic-program materials), then extracted and phrased representative natural-language questions whose answers are grounded in those documents. Each record retains the document-derived reference answer and source identifier, enabling reproducible retrieval and answer evaluation. The benchmark is therefore document-grounded rather than a log of production user

conversations or a survey of students; questions were curated for coverage of the target counseling intents.

The benchmark spans institutional categories including `actual_tuition`, `academic_rules`, `scholarship`, `student_loan`, `other`, `social_support`, `academic_program`, `exemption_policy`, and `exemption_basis`. Each record includes reference-answer text and source identifiers. We do not report a full-benchmark category distribution because the authoritative reports provide exact category counts only for the 100-question single-domain routing subset used in Scenario 3.

4.2 Experimental Tools and Infrastructure

The experimental testbed was deployed across dedicated hardware and cloud infrastructure: **Computational Hardware:** Local experiments and neural cross-encoder reranking were conducted on a GPU-accelerated workstation with an Intel Core i9 CPU, 64 GB RAM, and an NVIDIA GeForce RTX 4090 GPU (24 GB VRAM). **Software Stack:** CTU-CHAT is implemented in Python and FastAPI. Multi-agent state orchestration is executed via LangGraph under a StateGraph supervisor, with session state cached in Redis 7 (via `redis:7-alpine`). Relational curricula are hosted in Neo4j 5 with the APOC plugin. Document parent chunks (2,800 characters, 100-character overlap) are stored in PostgreSQL; child chunks (400 characters, 50-character overlap) are indexed in Qdrant using `bkai-foundation-models/vietnamese-bi-encoder (768-d)`. Neural candidate reranking employs `BAAI/bge-reranker-v2-m3`. **LLM Backbones and Cloud Evaluation:** Agent reasoning and response synthesis use `gemini-2.5-flash-lite` across all benchmark evaluations (with operational server support for `gemini-3.5-flash-lite`), while contextual query rewriting uses `gemini-3.1-flash-lite`. End-to-end generation and automated Ragas evaluation were executed via Google Cloud Vertex AI using service account authentication, employing `gemini-2.5-flash-lite` across 5 parallel worker threads. The 1,050 evaluations were completed in 7.7 minutes.

4.3 Evaluation Metrics

Multi-Agent Coordination Metrics (Scenario 3): Following isolated evaluation protocols, we assess: (1) **Agent Accuracy & Macro-F1:** proportion of queries routed to the target specialist (actual = expected) alongside macro-averaged F1 across the four specialist domains to reduce class frequency bias; (2) **Intent Accuracy:** accuracy of fine-grained intent classification; (3) **Tool Selection, Argument Exact Match (EM), and Result Accuracy:** whether the correct execution path (deterministic catalog lookup vs. typed LLM tool) is invoked, whether extracted arguments match ground truth with exact numeric and case-insensitive string equality, and whether numerical outputs are mathematically exact without forbidden tokens; (4) **End-to-End Pass Rate:** joint

success across tool selection, argument binding, and execution output; (5) **Robustness & Tool-Gating Accuracy**: correct tool triggering on complete inputs versus deliberate tool suppression on malformed or policy-only queries; (6) **Safe Result Behavior**: explicit abstention or diagnostic clarification without parameter fabrication or hallucination; and (7) **Bounded Completion Rate**: proportion of decisions terminating within the evaluated operational invariants ($\text{max_tool_calls} = 1, \text{timeout} = 60.0 \text{ s}$).

Retrieval Quality Metrics: Evaluated over top- k returned passages against gold document identifiers: (1) **Hit@ k** ($k \in \{1, 3\}$): proportion of queries where at least one gold document is retrieved within top- k : $\text{Hit@}k = \frac{1}{|Q|} \sum_{q \in Q} \mathbb{I}(\text{rank}(d_q^*) \leq k)$; (2) **Precision@7 (P@7)** and **Recall@7 (R@7)**: proportion of relevant passages in the common Top-7 context window, and proportion of gold passages retrieved; (3) **Mean Reciprocal Rank (MRR@10)**: reciprocal rank of the first relevant document: $\text{MRR} = \frac{1}{|Q|} \sum_{q \in Q} \frac{1}{\text{rank}(d_q^*)}$; and (4) **Latency (ms)**: retrieval turnaround time.

End-to-End Generation Metrics (Ragas): Evaluated using LLM-as-a-judge under Ragas [4]. All four scores lie in $[0, 1]$ (higher is better), but they measure different objects: (1) **Answer Relevancy (AR)**: how directly the generated answer addresses the user’s question, independent of whether every fact is correct; (2) **Context Recall (CR)**: how much of the factual information required by the reference answer is present in the retrieved Top-7 context; (3) **Context Precision (CP)**: how concentrated the retrieved Top-7 ranking is on relevant evidence, rewarding relevant passages appearing earlier; and (4) **Answer Correctness (AC)**: whether the generated answer is factually consistent with the ground-truth answer, including numerical values and cohort conditions. Thus AR/AC evaluate the answer, whereas CR/CP evaluate its supporting context; none of these four scores is a latency or retrieval-time measure.

4.4 Three Experimental Scenarios

To rigorously evaluate the system against our three research questions, Table 4 maps each research gap and question to its corresponding architectural component, experiment, and evaluation metrics.

Table 4. Systematic mapping between Research Gaps, Research Questions, Proposed Components, Experiments, and Empirical Metrics.

Gap	RQ	Proposed Component	Experiment	Target Metrics	Empirical Evidence
Gap 1	RQ1 (Multi-Agent)	Supervisor + 4 Specialists (C1)	Scenario 3	Agent Acc, Tool EM, Robustness	97.0% routing, 100% tool EM (tuition), 100% bounded
Gap 2	RQ2 (Allocation)	Graph vs. Hybrid Allocation (C2)	Scenario 2	Context Precision, Answer Correctness	T4 records the highest AC: 0.647
Gap 3	RQ3 (Routing & RAG)	Intent Routing + Stacking (C3)	Scenario 1	Hit@k, Recall@5, MRR@10, Latency	E5 attains 0.9000 Hit@1, 0.9600 Hit@3, 0.9327 MRR

Scenario 1 — Retrieval Component Stacking (E1–E5): Objective & RQ Alignment: Scenario 1 investigates **RQ3** and supports **RQ2** by evaluating the empirical contribution of progressively stacking retrieval components over the 150-case benchmark. *Configurations:* We compare five cumulative setups: (1) **E1 (BM25 Lexical)**: BM25 search over tokenized child chunks; (2) **E2 (Dense**

Semantic): 768-d vietnamese-bi-encoder vector search in Qdrant; (3) **E3 (Hybrid RRF):** BM25 and Dense fused via Reciprocal Rank Fusion ($k = 60$); (4) **E4 (Hybrid + Reranker):** E3 augmented with `bge-reranker-v2-m3` cross-attention; and (5) **E5 (Full Proposed System):** integrating IG-SRR subspace routing, Neo4j relational grounding, and parent-child chunk mapping. *Results & Interpretation:* As shown in Table 5, BM25 achieves an initial lexical baseline (0.6667 Hit@1) owing to rigid alphanumeric tokens (cohort IDs, decision numbers). Dense retrieval alone exhibits marked degradation (0.4267 Hit@1, 0.5115 MRR) because vector proximity conflates semantically similar regulations across different cohorts. Unranked hybrid fusion (E3) balances lexical and semantic representations (0.5067 Hit@1, 0.7606 Recall@5). Introducing cross-encoder reranking over hybrid candidates (E4) substantially improves lexical-semantic reordering (0.8000 Hit@1, 0.8650 MRR). Finally, full heterogeneous integration with IG-SRR subspace routing and Neo4j relational grounding (E5) achieves the highest performance across all retrieval metrics: **0.9000 Hit@1, 0.9600 Hit@3, 0.2627 Precision@5, 0.8656 Recall@5, and 0.9327 MRR@10**, outperforming the un-governed cross-reranked baseline E4 by +10.0% in Hit@1 and +6.77% in MRR@10.

Scenario 2 — End-to-End QA Generation and Modular Ablation (T1–T7): *Objective & RQ Alignment:* Scenario 2 addresses **RQ2** and **RQ3** by evaluating end-to-end answer synthesis and modular ablation across seven configurations on the 150 benchmark queries. *Configurations:* (T1) BM25+LLM; (T2) Dense+LLM; (T3) Hybrid+LLM; (T4) Proposed CTU-CHAT Full Stack; (T5) w/o Reranker; (T6) w/o Neo4j Knowledge Graph; and (T7) w/o Governance Filter (removing subspace routing). *Results & Interpretation:* All 1,050 responses were evaluated using Ragas LLM-as-a-judge on Vertex AI with the same Top-7 context cutoff for every configuration (Table 6). The full system (T4) improves over the hybrid baseline (T3) on AR (0.730→**0.781**), CR (0.758→0.795), CP (0.531→0.722), and AC (0.600→**0.647**). The reranker and governed Graph injection therefore improve answer quality under the evaluated setup, although the graph ablation (T6) obtains slightly higher CR/CP (0.798/0.739), so T4 does not dominate every context metric. Removing the governance filter (T7) reduces AR/CR/CP to 0.579/0.583/0.415, matching the dense-only failure pattern and yielding AC 0.482; this supports the value of subspace routing without claiming universal dominance. The dense baseline (T2) is similarly weak (AR 0.580, CR 0.583, CP 0.415, AC 0.482), while the no-reranker ablation (T5) retains AC 0.612 but lower AR and context scores than T4.

Scenario 3 — Multi-Agent Routing, Tool Reliability, and Robustness: *Objective & RQ Alignment:* Scenario 3 directly addresses **RQ1** by evaluating the multi-agent coordination, tool execution reliability, and bounded safety of CTU-CHAT independently from retrieval quality (Scenario 1) or generative answer synthesis (Scenario 2). It investigates three core empirical questions: (1) Does the Supervisor reliably route diverse student inquiries to the correct specialist

Table 5. Retrieval evaluation results of cumulative configurations (E1–E5) on the 150-question benchmark.

Configuration	Hit@1	Hit@3	Precision@5	Recall@5	MRR@10
E1 (BM25 Lexical)	0.6667	0.9000	0.2360	0.7783	0.7857
E2 (Dense Vector)	0.4267	0.5867	0.1493	0.5778	0.5115
E3 (Hybrid RRF)	0.5067	0.7933	0.2213	0.7606	0.6702
E4 (Hybrid + Cross-Reranker)	0.8000	0.9333	0.2450	0.8200	0.8650
E5 (Full Proposed System)	0.9000	0.9600	0.2627	0.8656	0.9327

Table 6. End-to-end QA generation and ablation evaluation results (T1–T7) on the 150-question benchmark using Ragas metrics with a common Top-7 context window.

Config.	Configuration Description / Ablation Target	AR	CR	CP	AC
T1	BM25 Lexical + LLM Baseline	0.737	0.790	0.653	0.596
T2	Dense Semantic + LLM Baseline	0.580	0.583	0.415	0.482
T3	Hybrid (BM25 + Dense RRF) + LLM Baseline	0.730	0.758	0.531	0.600
T4	Proposed System (CTU-Chat Full Stack)	0.781	0.795	0.722	0.647
T5	w/o Cross-Encoder Reranker (bge-reranker-v2-m3)	0.723	0.758	0.531	0.612
T6	w/o Neo4j Knowledge Graph Grounding	0.750	0.798	0.739	0.642
T7	w/o Governance Filter (No Subspace Routing)	0.579	0.583	0.415	0.482

Metric definitions:

AR—Answer Relevancy: whether the generated answer directly addresses the user’s question; this score alone does not establish factual correctness.

CR—Context Recall: whether the retrieved Top-7 context contains the information required to support the ground-truth answer.

CP—Context Precision: whether the retrieved passages are relevant, with greater reward when relevant evidence appears at higher ranks.

AC—Answer Correctness: factual agreement between the generated answer and the ground truth, including monetary amounts, cohorts, and eligibility conditions.

All scores are normalized to $[0, 1]$, and higher values indicate better performance.

and intent? (2) When quantitative calculation or catalog lookup is required, does the system accurately discriminate between deterministic structured lookups and typed LLM tool calls, extract arguments with exact match, and return mathematically exact outputs? (3) Under incomplete, invalid, or adversarial inputs attempting forced tool invocation, does the system safely suppress execution, request missing parameters, and terminate within bounded constraints?

Experimental Setup: Evaluated using `gemini-2.5-flash-lite` via Google Cloud Vertex AI (temperature = 0.0, timeout = 60.0s, maximum 1 tool call per decision, SDK retry 1, sequential execution with 1.0s delay, 3 repetitions). The evaluation spans three suites: (1) **Supervisor Routing (Panel A):** Evaluated on 100 single-domain queries from the benchmark ($N = 300$ runs for the LLM Supervisor, alongside $N = 100$ runs for a baseline rule-based regex router). Fifty composite cross-domain queries (CDICT*) combining curriculum requirements, academic standing, and tuition schedules were excluded from single-agent routing measurement because the supervisor routes each atomic turn to a single spe-

cialist; forcing composite queries into a single label like `actual_tuition` penalizes valid alternative specialist assignments (`academic_program`, `calculation`, or combined). The 100 queries span 9 source categories mapped to 4 specialists: `financial` (28 queries), `general` (49 queries), `scholarship` (14 queries), and `academic` (9 queries). (2) **Specialist Tool Reliability (Panel B)**: Evaluated across 60 production-tool cases ($N = 180$ runs across 3 repetitions) from `scenario3_production_tools.json`. The suite covers five cases for each of the 11 production tools (six academic graph tools, three financial graph/policy tools, two deterministic calculators) plus five no-tool cases. Graph-backed outputs additionally record whether the result came from Graph or from an empty/fallback path; fallback is not counted as a Graph hit. (3) **Adversarial Robustness and Failure Gating (Panel C)**: Evaluated across 20 adversarial failure scenarios ($N = 60$ runs across 3 repetitions) from `scenario3_robustness_cases.json`. These stress-test system boundaries against: missing mandatory arguments, out-of-range GPA/training points (DRL)/fee reduction percentages, policy questions where tool execution must be suppressed, cross-domain queries, tuition queries lacking admission cohort, Vietnamese decimal comma notation, context-free follow-up queries, and prompt injection attempts instructing the model to fabricate numerical parameters.

Table 7. Scenario 3: Multi-agent routing, specialist tool execution reliability, and robustness under a maximum of one tool call per evaluated decision (`max_tool_calls = 1`).

Panel A: Supervisor Routing and Specialist Selection ($N = 100$ queries; Rule: 1 rep / LLM: 3 reps, $N = 300$)							
Router Variant	Runs (N)	Agent Acc.	Macro-F1	Intent Acc.	Bounded p50 (ms)	p95 (ms)	
Rule-Based Router	100	85.0%	69.2%	69.0%	100.0%	0.0	0.1
LLM Supervisor (CTU-CHAT)	300	97.0%	97.8%	95.0%	100.0%	795.5	981.7
Panel B: Specialist Tool Execution Reliability ($N = 15$ runs per function, 3 reps)							
Target Function	Execution Type	Selection	Arg. EM	Result Acc.	E2E Pass	Bounded	Graph hit
<code>tra_cuu_nganh</code>	LLM Tool Call	100.0%	100.0%	100.0%	100.0%	100.0%	-
<code>so_san_hanh_nganh</code>	LLM Tool Call	100.0%	100.0%	100.0%	100.0%	100.0%	-
<code>tim_nganh</code>	LLM Tool Call	100.0%	100.0%	100.0%	100.0%	100.0%	-
<code>xem_chuoitien_quyet</code>	LLM Tool Call	100.0%	100.0%	100.0%	100.0%	100.0%	-
<code>mon_chung_gia_nganh</code>	LLM Tool Call	93.3%	93.3%	100.0%	93.3%	100.0%	-
<code>tim_nganh.co.mon</code>	LLM Tool Call	100.0%	100.0%	100.0%	100.0%	100.0%	-
<code>tra_cuu_hoc_phi_graph</code>	LLM Tool Call	80.0%	80.0%	100.0%	80.0%	100.0%	80.0%
<code>tra_cuu.co.so.mien.giam.graph</code>	LLM Tool Call	80.0%	80.0%	73.3%	73.3%	93.3%	78.6%
<code>tra_cuu.quy.dinh.hoc.phi</code>	LLM Tool Call	100.0%	100.0%	100.0%	100.0%	100.0%	40.0%
<code>tin_h_toan_hoc_phi</code>	LLM Tool Call	100.0%	80.0%	80.0%	80.0%	100.0%	-
<code>tin_h_tien_hoc_bong</code>	LLM Tool Call	100.0%	100.0%	100.0%	100.0%	100.0%	-
<code>no_tool</code>	No Tool Call	100.0%	100.0%	100.0%	100.0%	100.0%	-
Panel C: Robustness and Bounded Failure Handling ($N = 60$ runs under malformed/adversarial inputs)							
Adversarial Gate	Runs (N)	Agent Acc.	Tool Gate Safe	Result	E2E Pass	Bounded	Notes
Failure Gate Robustness	60	100.0%	100.0%	98.3%	98.3%	100.0%	-

Note: `tuition_lookup` evaluates the deterministic structured catalog path. Scholarship and tuition reduction evaluate typed LLM tool selection.

Results & Interpretation: (1) *Supervisor vs. Heuristic Routing:* The LLM Supervisor attains **97.0% Agent Accuracy**, **97.8% Macro-F1**, and **95.0% Intent Accuracy**, outperforming the rule router (85.0%, 69.2%, and 69.0%). The Macro-F1 result shows higher balance across the 4 specialist domains than the rule router. While the rule router achieves near-zero latency (0.1 ms), it performs worse on this routing set; the LLM Supervisor provides higher semantic

discernment with a sub-second p95 latency (981.7 ms, p50 795.5 ms). (2) *Specialist Tool Execution and Graph Coverage*: As shown in Panel B, five academic tools, tuition-policy lookup, scholarship calculation, and no-tool gating achieved **100.0%** end-to-end pass. The production graph tuition lookup achieved 80.0% selection/argument/E2E with an 80.0% Graph-hit rate. Graph exemption-basis lookup achieved 80.0% selection and 73.3% result/E2E, with a 78.6% Graph-hit rate. The only non-graph calculator weakness was 80.0% argument EM and E2E for tuition arithmetic. (3) *Robustness and Bounded Failure Invariants*: Under adversarial stress testing (Panel C), CTU-CHAT achieved **100.0% Tool/No-tool decision accuracy**, **100.0% Agent and Intent Accuracy**, **98.3% Safe Result behavior**, and **98.3% End-to-End pass**. The system suppressed tool calls on malformed or policy-only queries in this suite, explicitly requesting missing parameters rather than fabricating values. Across all 550 rescored isolated decision records in Panels A, B, and C, **100.0% completed within operational bounds** (max_tool_calls = 1, timeout = 60s), supporting bounded execution at the isolated decision level.

4.5 Discussion

Answering RQ1 (Domain-Specialized Multi-Agent Architecture): Centralizing routing under a structured supervisor with asymmetric privileges and bounds (max_tool_calls = 1, timeout = 60s) produced bounded isolated decisions in the Scenario 3 runs: the LLM Supervisor achieves 97.0% agent accuracy and 97.8% Macro-F1 across 300 evaluations, outperforming rule-based routing (85.0% and 69.2%). Across the 180 production-tool evaluations, academic lookup, policy lookup, scholarship, and no-tool gating reached 100.0% end-to-end pass, while graph tuition and exemption-basis paths exposed the remaining selection and backend-coverage failures. Adversarial testing demonstrated 100.0% tool-gating accuracy, 98.3% safe-result behavior, and 100.0% bounded completion.

Answering RQ2 (Heterogeneous Knowledge Allocation): Decomposing counseling data across relational curricula (Neo4j), tuition catalogs (JSON), and narrative decrees (hybrid RAG) is supported by the ablation pattern. Full heterogeneous integration with Neo4j grounding (E5) achieves 0.9000 Hit@1, 0.9600 Hit@3, 0.8656 Recall@5, and 0.9327 MRR. In the Top-7 end-to-end evaluation, CTU-CHAT (T4) attains the highest Answer Correctness (0.647) and improves substantially over the hybrid baseline (T3: 0.600), while the graph ablation (T6) is slightly higher on context recall/precision (0.798/0.739 versus T4's 0.795/0.722). Thus the measured benefit is strongest for answer correctness rather than universal context-metric dominance.

Answering RQ3 (Representation-Aware Evidence Routing): Across the cumulative configurations, top-1 retrieval increases from 0.6667 (E1) to **0.9000** (E5), with E5 achieving **0.9600 Hit@3**, **0.8656 Recall@5**, and **0.9327 MRR**. In the Top-7 ablation, T4's governed Graph injection and reranking raise AR/CR/CP/AC over T3 to 0.781/0.795/0.722/0.647, while T7 without governance falls to 0.579/0.583/0.415/0.482. Neural-reranked configurations also show higher measured latency (20.09 ms for E1, 36,274.77 ms for E4, and 59,011.74 ms

for E5), while E3 records 94.05 ms and may be considered when latency is prioritized.

4.6 Threats to Validity and Limitations

While derived from Can Tho University, the administrative taxonomy generalizes across Vietnamese higher education; multi-institutional validation remains planned. In addition, cross-attention reranking latency motivates model distillation. Curricula graphs currently rely on offline schemas, and automated judges penalize out-of-scope abstentions, underestimating practical correctness when refusing ungrounded generation. Finally, while Scenario 3 validates bounded execution and tool reliability at the isolated supervisor and specialist decision level, end-to-end fault injection across external infrastructure services (e.g., Neo4j connection drops or Qdrant timeouts) represents an orthogonal system-resilience evaluation reserved for future work.

5 Conclusion and Future Work

This paper addressed knowledge heterogeneity in university counseling, where uniform RAG pipelines and uncoordinated agent meshes can be brittle across relational curricula, structured tuition matrices, and narrative decrees. CTU-CHAT introduces a centralized supervisor-routed multi-agent architecture granting four domain specialists isolated subspaces, dedicated retrieval paths, and asymmetric tool privileges. Relational curricula are allocated to Neo4j and structured tuition to graph catalogs with deterministic engines, reserving hybrid RAG (BM25, dense embeddings, RRF, cross-encoder reranking) for narrative decrees. Specialists execute without peer-to-peer delegation, with at most one tool call per evaluated decision (`max_tool_calls = 1`).

Empirical evaluation provides evidence of 97.0% supervisor routing accuracy and 100.0% bounded completion in isolated Scenario 3 decisions. The production-tool suite achieved 100.0% end-to-end pass for academic lookup, tuition-policy, scholarship, and no-tool gates; graph tuition lookup reached 80.0%, while graph exemption-basis lookup reached 73.3% with a 78.6% Graph-hit rate. Component stacking achieves **0.9000 Hit@1**, **0.9600 Hit@3**, **0.8656 Recall@5**, and **0.9327 MRR**. With a common Top-7 context window, the full system T4 reaches the highest Answer Correctness (0.647), although the graph ablation T6 is slightly higher on context recall and precision; removing governance (T7) lowers AC to 0.482. Future work will explore cross-institutional transferability, latency distillation, and advisory trials.

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